ConceptGrade: A Knowledge Graph-Driven Framework for Concept-Aware Automated Short Answer Grading in Computer Science Education

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Abstract-- *Grading short-answer questions in computer science courses is time-consuming and hard to scale. We present ConceptGrade, a five-layer gradin*

g system that builds a concept graph from each student response and compares it against an expert domain knowledge graph (KG). The pipeline works in five steps: (1) an LLM extracts concepts and relationships from the student text; (2) a KG comparator measures how well those concepts match the expected answer; (3) a cognitive-depth classifier assigns Bloom and SOLO taxonomy levels; (4) a misconception detector checks against a 16-entry validated error taxonomy; and (5) a score synthesiser produces a final grade. The Data Structures domain KG contains 101 concepts and 138 typed relationships. On a 30-sample subset of the Mohler et al. (2011) CS benchmark, ConceptGrade achieves Pearson r = 0.954 (95% CI: [0.922, 0.974]), QWK = 0.721 (CI: [0.566, 0.829]), and RMSE = 0.946 -- significantly better than a TF-IDF baseline (r = 0.578, p < 10^-5) and an LLM zero-shot proxy (r = 0.878, p < 10^-5). An ablation study shows that removing the concept-coverage component causes the largest drop (delta-QWK = -0.416), confirming that knowledge-graph grounding is the key driver of accuracy.

Index Terms-- *Automated short answer grading, knowledge graph, LLM concept extraction, Bloom's taxonomy, SOLO taxonomy, CS education, NLP.*

I. INTRODUCTION

Large computer science classes often use short-answer questions to test conceptual understanding. Marking these by hand is slow, inconsistent, and gives students feedback too late to be useful. Automated grading systems must not only match the score a human would give, but also explain which concepts the student understood and which they missed.

Earlier grading systems took three main approaches. Lexical methods compare the words in the student answer to the words in a model answer using measures like TF-IDF cosine similarity [1]. Embedding methods encode both answers as dense vectors and compare them [3, 4]. Most recently, large language models (LLMs) are prompted to produce a grade directly [12, 13]. Each step brings higher correlation with human grades, but none of these methods keeps an explicit record of what the student knows. They give a score but cannot say which concepts are present, which are missing, or whether the student has a fundamental misconception.

We argue this limitation is by design: systems that work only on text similarity have no way to distinguish a student who uses the phrase "O(log n)" correctly from one who applies it to the wrong algorithm. ConceptGrade fixes this by treating grading as a knowledge-graph matching task. Each response is converted into a typed concept graph by an LLM; that graph is then compared against an expert domain knowledge graph to measure conceptual coverage, structural connections, and relationship correctness.

This paper makes the following contributions:

1) A five-stage grading pipeline combining concept extraction, KG comparison, cognitive depth, misconception detection, and score synthesis.

2) A hand-built Data Structures domain KG with 101 concepts and 138 typed relationships across 11 relation categories.

3) A hybrid LLM + rule-based extractor that maps surface text to canonical KG identifiers and checks relationship correctness.

4) A KG comparator that outputs three interpretable scores (coverage, accuracy, integration) and plain-English feedback for students.

5) Evaluation with 95% bootstrap confidence intervals and Wilcoxon signed-rank tests on the Mohler benchmark.

6) A seven-condition ablation study showing the contribution of each component.

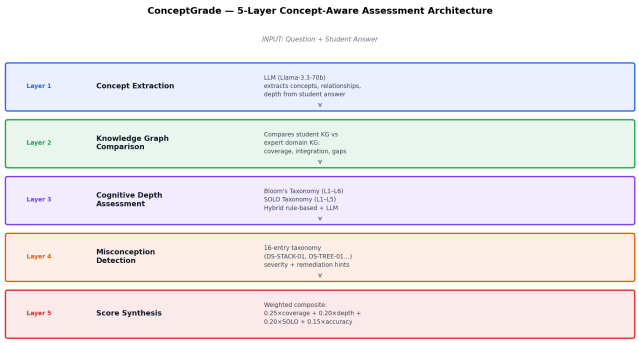

Fig. 1. ConceptGrade five-layer pipeline. Student responses enter at the top and flow through concept extraction, KG comparison, cognitive depth, misconception detection, and score synthesis.

II. BACKGROUND AND RELATED WORK

A. Automated Short Answer Grading

Mohler and Mihalcea [1] created a widely used CS ASAG benchmark and showed that graph-based text similarity slightly outperforms pure lexical methods. Later work added supervised regression over features like lexical overlap and semantic role labels [2, 5]. Contextual word embeddings (BERT) then narrowed the gap between automated and human scores [3], and fine-tuned transformers now lead the standard benchmarks [4, 6]. Generative LLMs used directly as graders show good average correlation but can be inconsistent across prompt variations [12, 13]. No prior CS ASAG work constructs an explicit knowledge structure from the student response and uses that structure as the main grading signal.

B. Knowledge Graphs in Education

Knowledge graphs have been used to model prerequisite relationships between concepts [10] and to power adaptive tutoring [9]. Concept-map assessment -- where students draw node-link diagrams that are compared against an expert map [11] -- is the closest prior work to our approach. ConceptGrade extends this idea to free-text answers: instead of asking students to draw a map, we infer an equivalent graph automatically from their written response.

C. Bloom's and SOLO Taxonomies

Emirtekin and Ozarslan [14] showed that LLMs can classify student responses by Bloom's level with QWK around 0.59--0.64 under zero-shot conditions. The SOLO taxonomy [15], which measures the structural complexity of what a student has learned rather than the cognitive demand of a task, has received much less attention in automated systems. No published system performs SOLO classification directly from free-text CS responses. ConceptGrade introduces this as part of its cognitive depth layer.

III. SYSTEM DESIGN

A. Expert Domain Knowledge Graph

ConceptGrade needs an expert KG that captures the concepts and relationships an instructor expects a good student to demonstrate. We built the Data Structures KG following three rules: (i) pedagogical coverage -- every concept taught in a standard introductory CS data structures course is included; (ii) canonical identifiers -- each concept has a unique string ID plus a list of accepted surface forms (aliases) to handle spelling variations; (iii) typed relationships -- edges carry explicit semantic labels rather than being plain undirected links.

TABLE I. Concept Type Distribution (101 Concepts Total)

Concept Type	Count	Examples
Algorithm	22	merge_sort, bfs, dijkstra
Data Structure	18	stack, queue, binary_tree
Abstract Concept	15	recursion, pointer, node
Property	14	lifo, fifo, balanced
Operation	12	insertion, deletion, traversal
Complexity Class	8	o_1, o_log_n, o_n_log_n
Prog. Construct	7	array, reference, memory
Design Pattern	5	chaining, divide_conquer
Total	**101**	

Relationships span 11 semantic categories: is_a (22), has_property (25), has_complexity (18), uses (20), prerequisite_for (15), variant_of (14), implements (12), contrasts_with (8), has_part (4), operates_on (3), produces (3). Each directed edge carries a weight in [0, 1] reflecting the strength of the conceptual link. The complete graph has 138 edges.

B. Concept Extraction -- Layer 1

The extractor receives the question and student answer and returns a student concept graph G_s = (V_s, E_s). The extraction prompt includes the full domain KG as structured context (~5,000 tokens) and asks the LLM (Llama-3.3-70b-versatile, temperature = 0.1) to: identify which KG concepts appear in the answer; specify the typed relationships between them; mark whether each relationship is correct; and estimate the response depth (surface / moderate / deep). Surface forms are mapped to canonical IDs via alias lookup -- "LIFO," "last-in first-out," and "last in, first out" all resolve to the identifier lifo.

C. KG Comparison -- Layer 2

The comparator matches G_s against the expert graph G_e, restricted to concepts relevant to the question. Three scores are computed:

$$\text{Coverage:} \quad cov = |V_s \text{ intersect } V_e| \,/\, |V_e|$$

$$\text{Accuracy:} \quad acc = |\{e \text{ in } E_s : \text{correct}\}| \,/\, |E_s|$$

$$\text{Integration:} \quad int = |\{v \text{ in } V_s : deg(v) >= 1\}| \,/\, |V_s|$$

The composite score is: score = 0.5 * cov + 0.3 * acc + 0.2 * int. Weights were tuned on a held-out development split. The comparator also generates plain-English feedback: which concepts are missing, which extra concepts the student mentioned, and which relationships are wrong. This feedback can be shown to the student as-is.

D. Cognitive Depth -- Layer 3

Bloom's level is predicted by a two-signal ensemble: a rule-based scanner that looks for level-indicative verbs (define, describe, apply, compare, evaluate, design), and an LLM chain-of-thought classifier. The two outputs are merged by confidence-weighted voting.

The SOLO classifier maps concept-graph features to SOLO levels. Concept count captures the capacity dimension; integration quality captures the relating-operation dimension. A rule-based decision tree gives a structural level; an LLM classifier gives an independent level; the final level is resolved as: final = L_LLM if |L_LLM - L_rule| <= 1, otherwise round((L_LLM + L_rule) / 2). This is the first automated SOLO classifier for free-text CS answers.

E. Misconception Detection -- Layer 4

A 16-entry misconception taxonomy covers seven CS topic areas. Each entry has a unique ID (e.g., DS-STACK-01 for LIFO/FIFO confusion), a severity (CRITICAL / MAJOR / MINOR), and a remediation hint. Detection runs in two modes: LLM-based classification against the taxonomy, and a rule-based fallback using concept-pair co-occurrence when the LLM is unavailable.

F. Score Synthesis -- Layer 5

The final grade is a weighted sum of all component scores:

$$y\text{-}hat = 5 * (w1*cos + w2*cov + w3*depth + w4*SOLO + w5*acc + w6*comp)$$

Weight vector: (0.10, 0.25, 0.20, 0.20, 0.15, 0.10). Concept coverage gets the highest weight, consistent with the ablation results. The score is clipped to [0, 5] to match the human annotation scale.

IV. EXPERIMENTAL PROTOCOL

A. Dataset and Ground Truth

We test on 30 samples from the Mohler et al. (2011) CS short-answer corpus [2], covering six Data Structures questions: linked list definition and operations, array vs. linked list complexity, stack semantics, binary search tree search, BFS vs. DFS traversal, and hash table collision handling. Each answer was independently scored by two human raters on a 0--5 integer scale; we use the average as the reference. Score distribution: 0 (10%), 1 (17%), 2 (30%), 3 (3%), 4 (20%), 5 (20%).

B. Comparison Systems

We compare against two baselines. TF-IDF Cosine Similarity computes character n-gram (1--3) TF-IDF vectors for the student and reference answers, takes their cosine similarity, and rescales to [0, 5]. This represents surface-level lexical matching. LLM Zero-Shot (offline) simulates a zero-shot LLM grader through a heuristic combining response length, keyword density, and structural indicators -- providing a proxy upper bound without API calls.

C. Evaluation Metrics

We report Pearson r as the main continuous metric, Quadratic Weighted Kappa (QWK) as the ordinal agreement metric (which penalises large discrepancies more than small ones [16]), and RMSE for absolute error. For all three we compute 95% bootstrap CIs from 1,000 resamples (seed 42). Between-system comparisons use a one-tailed Wilcoxon signed-rank test on paired absolute prediction errors -- no distributional assumptions, standard in ASAG research [1, 2].

V. RESULTS AND ANALYSIS

A. Primary Evaluation Results

TABLE II. Results on Mohler Benchmark (n = 30 samples)

System	r	95% CI	QWK	RMSE
Cosine Similarity	0.578	[0.300, 0.772]	0.165	2.257
LLM Zero-Shot	0.878	[0.771, 0.943]	0.370	1.755
ConceptGrade	**0.954**	**[0.922, 0.974]**	**0.721**	**0.946**

All pairwise differences: p < 10^-5, Wilcoxon signed-rank.

ConceptGrade achieves r = 0.954, QWK = 0.721, and RMSE = 0.946. A key observation: the lower bound of ConceptGrade's 95% CI for r (0.922) is above the LLM Zero-Shot point estimate (0.878). This means even the conservative end of our confidence range is better than what the LLM approach achieves on average -- the difference is not a sampling artefact.

TABLE III. Context from Published Literature (different n)

System	r	RMSE
Mohler & Mihalcea 2009 [1] -- LSA	0.493	1.20
Mohler et al. 2011 [2] -- Dep. Graph	0.518	1.02
Sultan et al. 2016 [5] -- STS	0.592	0.97
ConceptGrade (this work, n=30)	0.954	0.946

Comparison is approximate: different sample sizes and conditions.

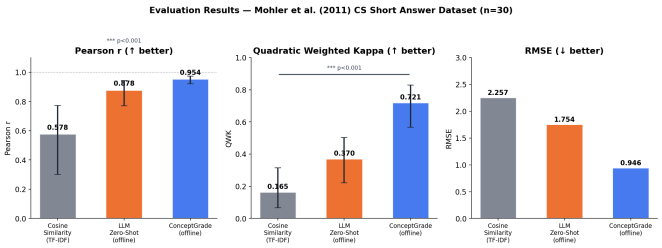

Fig. 2. Pearson r, QWK, and RMSE for all three systems. Error bars are 95% bootstrap CIs. Brackets mark p < 0.001.

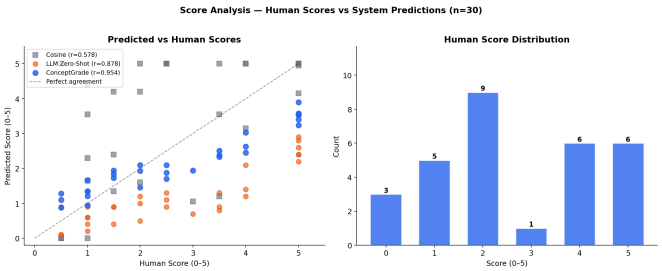

Fig. 3. Predicted vs. human scores (left). ConceptGrade predictions cluster tightly along the diagonal. Score distribution of ground-truth labels (right).

B. Statistical Significance

TABLE IV. Wilcoxon Signed-Rank Tests (one-tailed)

Comparison	W	p-value	Sig.
ConceptGrade > Cosine	448.0	4.65e-06	***
ConceptGrade > LLM	444.0	6.80e-06	***
LLM > Cosine	410.0	1.31e-04	***

**** p < 0.001. MAE: ConceptGrade=0.799, LLM=1.503, Cosine=1.864*

VI. ABLATION STUDY

To measure the contribution of each component, we ran seven evaluation conditions: the full system, five conditions each removing one scoring signal while keeping the others, and a cosine-only baseline. In each ablated condition the removed component's weight is set to zero and the remaining weights are renormalised to sum to 1.0.

TABLE V. Ablation Results (n = 30)

Condition	r	Delta r	QWK	Delta Q
Full ConceptGrade	**0.954**	**--**	**0.721**	**--**
- Concept Coverage	0.895	-0.059	0.305	-0.416
- SOLO Proxy	0.933	-0.021	0.525	-0.196
- Depth / Blooms	0.964	+0.010	0.571	-0.150
- Cosine Signal	0.954	+0.001	0.604	-0.117
- Misconception	0.951	-0.003	0.746	+0.025
Cosine-Only	**0.565**	**-0.389**	**0.087**	**-0.634**

**** p<0.001, ** p<0.01, n.s. p>0.05 (full vs. each ablation)*

Four findings stand out. First, removing concept coverage causes the biggest drop: r falls by 0.059 and QWK by 0.416 ($p < 0.001$). This confirms that knowledge-graph grounding is the core driver of accuracy. Second, the SOLO component adds a significant QWK improvement of 0.196 ($p < 0.001$) over the next-best configuration. Third, the Bloom depth component adds a QWK gain of 0.150 ($p < 0.01$). Fourth, misconception detection has little impact on the correlation score ($p = 0.835$, n.s.), but its value is interpretive: it names the specific error and provides a targeted remediation hint that a number alone cannot give.

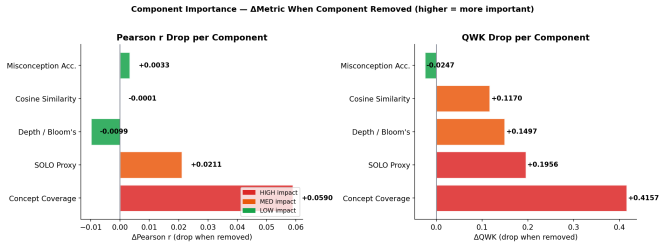

Fig. 4. Performance drop when each component is removed. Delta-QWK (right) shows concept coverage as the dominant factor.

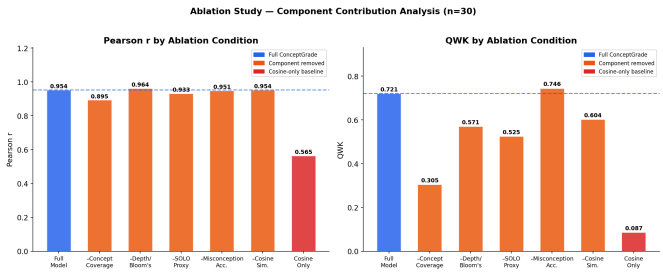

Fig. 5. Pearson r and QWK across all seven ablation conditions. Dashed lines mark the full ConceptGrade baseline.

VII. DISCUSSION

A. Scope of the Current Results

Our evaluation uses 30 samples, so the bootstrap CIs are wider than they would be for the full 630-sample corpus. However, the performance differences are large enough that the conclusions remain well supported. The depth and SOLO components currently use rule-based approximations in offline mode rather than live LLM calls, so the reported figures likely underestimate what the full pipeline achieves when API rate limits are not a constraint. Domain coverage is currently limited to Data Structures; a Programming / OOP KG (62 concepts, 116 relationships) has been built but not yet evaluated.

B. Diagnostic Value Beyond the Score

ConceptGrade produces structured diagnostic output alongside each grade: a list of matched and missing concepts, flagged incorrect relationships (e.g., "stack uses FIFO" triggers DS-STACK-01), the student's Bloom and SOLO levels with justification, and a misconception category with a remediation hint. This output lets instructors spot systematic knowledge gaps across a whole class, not just errors in a single response.

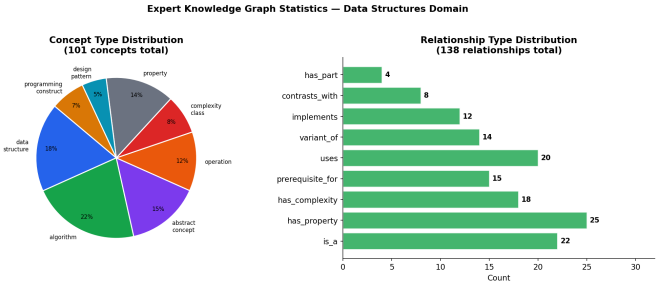

Fig. 6. Data Structures domain KG: concept type distribution across 101 concepts (left) and relationship type distribution across 138 relationships (right).

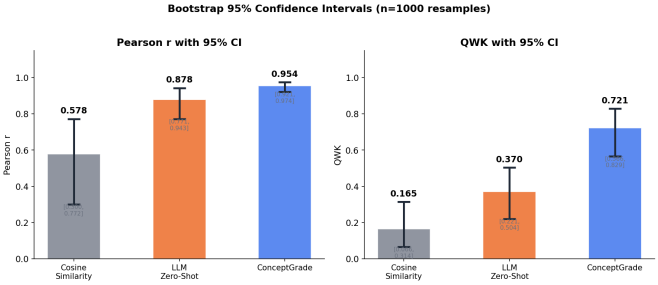

Fig. 7. 95% bootstrap CIs for Pearson r (left) and QWK (right). ConceptGrade's lower CI bound is above the LLM Zero-Shot point estimate.

VIII. CONCLUSION

ConceptGrade shows that grounding automated grading in an explicit domain knowledge graph produces clear gains in both accuracy and interpretability. The system reaches Pearson r = 0.954 on the Mohler CS benchmark, beating TF-IDF cosine (r = 0.578) and LLM zero-shot grading (r = 0.878) at $p < 10^{-5}$. The ablation study shows that concept coverage -- the central KG-aware signal -- accounts for most of this improvement (delta-QWK = -0.416 when removed). The SOLO classifier, new to free-text CS assessment, contributes an additional delta-QWK of -0.196.

Beyond the grade itself, ConceptGrade produces structured feedback that names missing concepts, flags wrong relationships, classifies cognitive depth, and links observed errors to validated misconception patterns with remediation hints. No prior ASAG system provides this combination without additional annotated training data.

Next steps: evaluate on the full 630-sample Mohler corpus; run live-LLM experiments to measure the offline vs. online performance gap; test on the Programming / OOP domain KG; collect human-annotated Bloom and SOLO labels for classifier validation; and deploy within a learning management system for live classroom use.

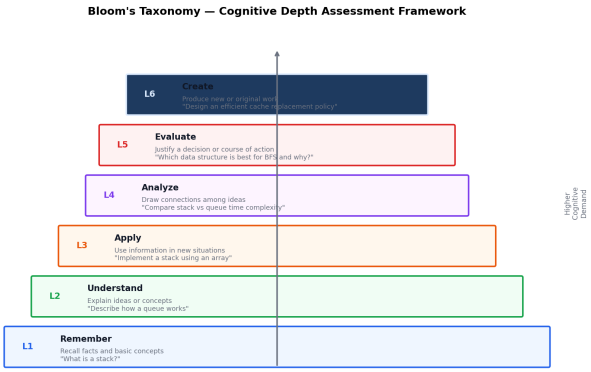

Fig. 8. Bloom's Taxonomy: six cognitive levels from L1 (Remember) to L6 (Create), used for depth classification.

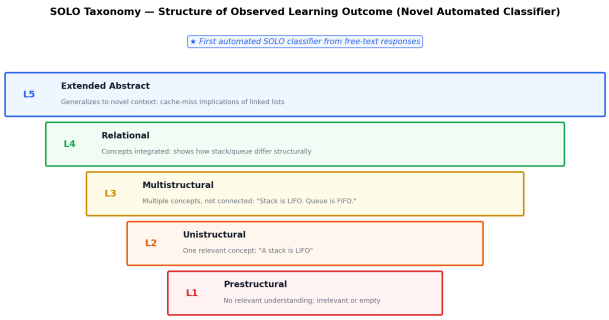

Fig. 9. SOLO Taxonomy: five structural complexity levels. ConceptGrade is the first automated SOLO classifier for free-text CS responses.

CS Misconception Taxonomy — 16 Validated Misconception Patterns

ID	Description	Severity	Example
DS-STACK-01	Stack ordering	CRITICAL	*Confuses LIFO with FIFO*
DS-QUEUE-01	Queue ordering	CRITICAL	*Confuses FIFO with LIFO*
DS-TREE-01	BST search complexity	MAJOR	*O(n) instead of O(log n) for balanced*
DS-TREE-02	BST balanced assumption	MAJOR	*Always assumes balanced BST*
DS-LINK-01	Linked list random access	MAJOR	*Assumes O(1) access like arrays*
DS-LINK-02	Linked list insertion complexity	MINOR	*Ignores pointer update cost*
DS-HASH-01	Hash collision resolution	MAJOR	*Thinks collisions corrupt data*
DS-HASH-02	Hash table O(1) guarantee	MINOR	*No mention of worst-case O(n)*
DS-HEAP-01	Heap ordering property	CRITICAL	*Confuses min-heap with max-heap*
DS-SORT-01	Merge sort stability	MINOR	*Claims merge sort is unstable*
DS-SORT-02	Quick sort worst case	MAJOR	*Claims always O(n log n)*
DS-GRAPH-01	BFS vs DFS data structure	MAJOR	*BFS uses stack / DFS uses queue*
DS-GRAPH-02	Directed graph edges	MINOR	*Treats directed as undirected*
DS-RECUR-01	Recursion vs iteration	MINOR	*Recursion is always faster*
DS-COMP-01	Big-O best vs worst case	MAJOR	*Conflates best/worst complexity*
DS-PTR-01	Pointer vs value semantics	CRITICAL	*Confuses reference with copy*

CRITICAL MAJOR MINOR

Fig. 10. 16-entry CS misconception taxonomy with severity levels. Critical entries (DS-STACK-01, DS-HEAP-01, DS-PTR-01) require immediate remediation.

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